

WEB-BASED FOOD ORDERING SYSTEM

Group Project Design Document

SE2030 - Software Engineering

1. INTRODUCTION

This document provides the comprehensive design specification for the Web-Based Food Ordering System. The system is divided into six major functional modules, each assigned to a group member. This document ensures cohesive integration and clear specifications for all development activities.

2. PROJECT OVERVIEW

The Web-Based Food Ordering System is a comprehensive platform that enables customers to browse restaurant menus, place orders, track delivery, and manage payments through an intuitive web interface. The system is divided into six independent modules that work together to provide a complete food ordering solution.

2.1 Project Objectives

- Enable seamless food ordering experience for customers
- Provide restaurant owners with menu management capabilities
- Facilitate real-time order tracking and delivery management
- Ensure secure payment processing
- Optimize kitchen workflow through order status management

3. SYSTEM ARCHITECTURE

The system follows a modular architecture where each major functional module is independently designed with clear interfaces for communication with other modules. This allows team members to work independently while maintaining system integration.

3.1 Module Overview

Member	Module	Primary User	Key Responsibility
Member 1	Customer Registration & Profile	Customer	User authentication and profile management
Member 2	Menu Management	Admin/Restaurant Owner	Menu item and category management
Member 3	Order Placement & Cart	Customer	Shopping cart and order placement

Member 4	Payment Processing	Customer	Payment processing and invoicing
Member 5	Order Tracking & Status	Customer/Kitchen	Order status tracking and updates
Member 6	Delivery Management	Delivery Person/Admin	Delivery assignment and tracking

4. DETAILED MODULE SPECIFICATIONS

4.1 MODULE 1: Customer Registration & Profile Management

Member: Member 1

Primary User: Customer

Description: This module handles all customer account-related activities including registration, login, profile updates, and viewing order history.

Key Responsibilities: Register new customer account, secure login/logout, view and update personal profile information, manage saved delivery addresses, view past order history

4.2 MODULE 2: Menu

ManagementMember: Member 2

Primary User: Admin / Restaurant Owner

Description: This module allows the admin to manage the restaurant's digital menu including food items, categories, pricing, and availability.

Key Responsibilities: Add new food items with name, description, price, and image; Organize items into categories; Edit existing food item details; Mark items as available or unavailable; Delete food items

4.3 MODULE 3: Order Placement & Cart Management

Member: Member 3

Primary User: Customer

Description: This module handles the complete ordering process from browsing the menu, managing the cart, applying discounts, and confirming orders.

Key Responsibilities: Browse menu items and view details; Add items to shopping cart; Update item quantities or remove items from cart; Apply promo codes or discounts; Review order summary and place the order

4.4 MODULE 4: Payment Processing

Member: Member 4

Primary User: Customer

Description: This module manages the payment flow after an order is placed, including payment method selection, confirmation, invoice generation, and payment history.

Key Responsibilities: Select payment method (Cash on Delivery, Card, Online); Enter and validate payment details; Process and confirm payment; Generate order receipt/invoice; View payment history

4.5 MODULE 5: Order Tracking & Status Management

Member: Member 5

Primary User: Customer & Kitchen Staff

Description: This module enables real-time order tracking for customers and allows kitchen staff to update order statuses as they progress through preparation stages.

Key Responsibilities: Kitchen staff updates order status at each stage; Customer views live order status; Admin monitors all active orders; Notify customer on each status change; View full order status history

4.6 MODULE 6: Delivery

ManagementMember: Member 6

Primary User: Delivery Person & Admin

Description: This module manages the delivery process including assigning delivery personnel to orders, tracking deliveries, and confirming successful delivery.

Key Responsibilities: Admin assigns orders to available delivery persons; Delivery person views their assigned deliveries; Update delivery status at each stage; Confirm delivery completion; Admin monitors all ongoing deliveries

5. DESIGN PATTERNS

This project applies industry-standard design patterns to ensure maintainability, scalability, and loose coupling between components. The following patterns are used:

5.1 Singleton Pattern

Purpose: Ensure only one instance of critical system classes

Application: Database Connection Manager, Configuration Manager, Payment Gateway

Implementation: Private constructor with static getInstance() method

5.2 Observer Pattern

Purpose: Enable real-time notifications for order status changes

Application: Order Status System notifies Kitchen Staff, Delivery Personnel, and Customers

Benefits: Loose coupling, dynamic subscription management, automatic updates

5.3 Factory Pattern

Purpose: Create different payment method objects

Application: PaymentFactory creates Cash, Card, or Online payment objects

Benefits: Extensible payment methods, reduces client complexity

6. CRUD OPERATIONS SPECIFICATION

All modules follow consistent CRUD (Create, Read, Update, Delete) operation patterns to ensure predictable behavior and maintainability. The following table outlines the standard CRUD operations:

Operation	Description	Expected Outcome	Database Impact
CREATE	Insert new entity data	Success confirmation, new record visible immediately	New row inserted, transaction committed
READ	Retrieve entity data	Data returned accurately, searchable/filterable	No changes, query executed
UPDATE	Modify existing entity	Changes saved and reflected immediately	Record updated, timestamp refreshed
DELETE	Remove entity record	Record removed from UI, no longer searchable	Row deleted or soft-deleted (status marked)

6.1 MODULE 1 - Customer Registration & Profile Management

CREATE: Register a new customer account with name, email, password, phone. Add a new delivery address to the profile.

READ: View personal profile details (name, email, phone, address). View order history list with order dates and statuses.

UPDATE: Edit profile details (name, phone, address). Change account password securely.

DELETE: Remove a saved delivery address. Delete customer account permanently.

6.2 MODULE 2 - Menu Management

CREATE: Add a new food item with name, description, price, category, and image. Create a new food category (e.g., Beverages, Starters).

READ: View all food items with details and availability status. Filter and search items by category or name.

UPDATE: Edit food item details (name, price, description, image). Toggle item availability (In Stock / Out of Stock).

DELETE: Delete a food item from the menu. Delete an empty food category.

6.3 MODULE 3 - Order Placement & Cart Management

CREATE: Add a food item to the cart with selected quantity. Place a confirmed order from the cart.

READ: View current cart contents with item names, quantities, and prices. Browse menu items with names, images, and prices.

UPDATE: Update item quantity in the cart (increase/decrease). Apply a promo code to get discount on order.

DELETE: Remove a specific item from the cart. Clear the entire cart before placing the order.

6.4 MODULE 4 - Payment Processing

CREATE: Submit payment for a placed order with selected payment method. Generate and save a payment receipt/invoice for the order.

READ: View payment details for a specific order. View full payment history with dates and amounts.

UPDATE: Update payment status (Pending → Paid) after verification. Update payment method before final confirmation.

DELETE: Cancel a pending payment record if order is cancelled. Remove payment history records (admin only).

6.5 MODULE 5 - Order Tracking & Status Management

CREATE: Create an initial order status entry when order is placed (status: Received). Add a status update log entry at each stage change.

READ: Customer views current order status and progress. Admin/Kitchen views all active orders and their statuses.

UPDATE: Kitchen updates order status (Received → Preparing → Ready → Out for Delivery → Delivered). Admin manually updates order status if needed.

DELETE: Remove completed/old order status records after archiving. Cancel an order and update status to Cancelled.

6.6 MODULE 6 - Delivery Management

CREATE: Create a delivery record when an order is assigned to a delivery person. Register a new delivery person account in the system.

READ: Delivery person views their list of assigned deliveries. Admin views all deliveries with status and assigned person.

UPDATE: Update delivery status (Assigned → Picked Up → On the Way → Delivered). Reassign delivery to a different person if needed.

DELETE: Remove a delivery record when order is cancelled. Remove delivery person account from system.

7. DATABASE DESIGN

The database schema is normalized to Third Normal Form (3NF) and supports all CRUD operations across all modules.

7.1 Key Entities

- **Customer:** Stores user account and profile information
- **MenuItem:** Represents food items in menu
- **Order:** Customer order containing multiple items
- **OrderItem:** Individual items in an order
- **Payment:** Payment transaction records
- **OrderStatus:** Status history and current state
- **Delivery:** Delivery assignment and tracking

8. IMPLEMENTATION GUIDELINES

8.1 Development Standards

- Use consistent naming conventions across all modules
- Implement interfaces for all major class contracts
- Apply error handling and logging consistently
- Use dependency injection for loose coupling
- Document all public methods and classes

8.2 Integration Points

- Modules communicate through well-defined interfaces
- Use event-driven communication between modules
- Share database access through centralized connection manager
- Implement consistent error codes and messages

8.3 Testing Strategy

- Unit test individual module classes
- Integration test module interactions
- Test all CRUD operations for each module
- Perform system testing for end-to-end workflows

9. CONCLUSION

This design document provides a comprehensive blueprint for implementing the Web-Based Food Ordering System. By adhering to these specifications and design patterns, the development team can ensure a scalable, maintainable, and robust system that meets all functional requirements.

The modular architecture allows each team member to work independently while maintaining integration cohesion through well-defined interfaces. Regular code reviews and integration testing will ensure the quality of the final product.